Random holdout

Temporal holdout

A

B

*All substitutions*

antiGen (multi−tree) 2.39
antiGen (single−tree) 2.23
Where x which 1.76
PSSM (sequence composition) 1.67
Frequency (memorisation) 1.55
Substitution matrix 1.41
Codon accessibility 0.77
Where only 0.41
Uniform 0.00

antiGen (multi−tree) 2.04
antiGen (single−tree) 1.78
Where x which 1.61
PSSM (sequence composition) 0.87
Frequency (memorisation) 1.34
Substitution matrix 1.29
Codon accessibility 0.67
Where only 0.38
Uniform 0.00

0 1 2

Nats better than uniform

C

D

*De novo only*

antiGen (multi−tree) 0.03
antiGen (single−tree) −0.57
Where x which 0.74
PSSM (sequence composition) 0.11
Frequency (memorisation) −0.23
Substitution matrix 0.76
Codon accessibility 0.70
Where only 0.03
Uniform 0.00

antiGen (multi−tree) 0.11
antiGen (single−tree) −0.68
Where x which 0.85
PSSM (sequence composition) −0.59
Frequency (memorisation) −0.22
Substitution matrix 0.76
Codon accessibility 0.43
Where only 0.08
Uniform 0.00

0 1 2

Nats better than uniform

All substitutions  De novo only  95% interval excludes zero  interval includes zero

E

F

*Versus single−tree*

SARS−CoV−2 S
H5Nx HA
Measles
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
**Pooled**

−3 −2 −1 0

−2.0 −1.5 −1.0 −0.5 0.0 0.5

Multi−tree minus comparator, NLL per mutation

G

H

*Versus where x which*

SARS−CoV−2 S
H5Nx HA
Measles
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
**Pooled**

−2 0 2

−1 0 1

Multi−tree minus comparator, NLL per mutation

Pooled−validation epoch (deployed)  Per−tree best epoch

I

J

*NLL per mutation*  *Recall at top 10%*

*Epoch rule*

single−tree model All substitutions
single−tree model De novo only
where x which All substitutions
where x which De novo only

−0.5 0.0 0.5 1.0 −0.05 0.00 0.05 0.10

−1.0 −0.5 0.0 0.5 1.0 −0.1 0.0 0.1 0.2

Pooled estimate